

Operating Systems (OS)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 55

Question 1 of 55 | GATE 2025 Set 1 (CS1) | Q.14 (Computer Science & IT)

Q.14

Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?

- (A) 221
- (B) 220
- (C) 222
- (D) 224

Question 2 of 55 | GATE 2025 Set 1 (CS1) | Q.29 (Computer Science & IT)

Q.29

Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer)

```
int main()
{
    int x=0,i=0;
    scanf("%d",&x);
    for(i=0; i<20; i++)
    {
        x = x+20;
        printf("%d\n",x);
    }
    return 0;
}
```

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 3 of 55 | GATE 2025 Set 1 (CS1) | Q.38 (Computer Science & IT)

Q.38

A computer has two processors, M1 and M2. Four processes P1, P2, P3, P4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows:

- M1 uses priority of execution for the processes as, $P1 > P3 > P2 > P4$, i.e., P1 and P4 have highest and lowest priorities, respectively.
- M2 uses priority of execution for the processes as, $P2 > P3 > P4 > P1$, i.e., P2 and P1 have highest and lowest priorities, respectively.

A process P_i is scheduled to a processor M_k , if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?

- (A) 9.00
- (B) 8.75
- (C) 6.50
- (D) 7.50

Question 4 of 55 | GATE 2025 Set 1 (CS1) | Q.53 (Computer Science & IT)

Q.53

A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. Hit rates and access times of various memory units are shown in the figure. The average memory access time in nanoseconds (ns) is _____. (rounded off to two decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 5 of 55 | GATE 2025 Set 1 (CS1) | Q.54 (Computer Science & IT)

Q.54

In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows:

The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.

A process accesses the pages in the following order of page numbers:

1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4.

If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____. (Answer in

integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 6 of 55 | GATE 2025 Set 2 (CS2) | Q.26 (Computer Science & IT)

Q.26

Processes P1, P2, P3, P4 arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times.

Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?

- (A) 22
- (B) 15
- (C) 37
- (D) 19

Question 7 of 55 | GATE 2025 Set 2 (CS2) | Q.47 (Computer Science & IT)

Q.47

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

initially

after

-

1* 2* 3* 4* 5* 4 1 6* 4 5 1* 3* 2*

-

1

1

1

1

1

1 1

6

6 6

6

6

2

-

-

2

2
4
4
4 4
4
4 4
1
1
1
-
-
-
3
3
5
5 5
5
5 5
5
3
3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

- (A) Least Recently Used page replacement policy
- (B) Least Frequently Used page replacement policy
- (C) Most Frequently Used page replacement policy
- (D) Optimal page replacement policy

Question 8 of 55 | GATE 2025 Set 2 (CS2) | Q.48 (Computer Science & IT)

Q.48

$P = \{P1, P2, P3, P4\}$ consists of all active processes in an operating system.

$R = \{R1, R2, R3, R4\}$ consists of single instances of distinct types of resources in the system.

The resource allocation graph has the following assignment and claim edges.

Assignment edges: $R1 \rightarrow P1, R2 \rightarrow P2, R3 \rightarrow P3, R4 \rightarrow P4$ (the assignment edge

$R1 \rightarrow P1$ means resource $R1$ is assigned to process $P1$, and so on for others)

Claim edges: $P1 \rightarrow R2, P2 \rightarrow R3, P3 \rightarrow R1, P2 \rightarrow R4, P4 \rightarrow R2$ (the claim edge

$P1 \rightarrow R2$ means process $P1$ is waiting for resource $R2$, and so on for others)

Which of the following statement(s) is/are CORRECT?

- (A) Aborting P1 makes the system deadlock free.
- (B) Aborting P3 makes the system deadlock free.
- (C) Aborting P2 makes the system deadlock free.
- (D) Aborting P1 and P4 makes the system deadlock free.

Question 9 of 55 | GATE 2025 Set 2 (CS2) | Q.58 (Computer Science & IT)

Q.58

A computer system supports a logical address space of 232 bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size.

The value of b is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 10 of 55 | GATE 2024 Set 1 (CS1 S5) | Q.24 (Computer Science & IT)

Q.24

Which of the following statements about threads is/are TRUE?

- (A) Threads can only be implemented in kernel space
 - (B) Each thread has its own file descriptor table for open files
 - (C) All the threads belonging to a process share a common stack
 - (D) Threads belonging to a process are by default not protected from each other
- Organizing Institute: IISc Bengaluru

Question 11 of 55 | GATE 2024 Set 1 (CS1 S5) | Q.53 (Computer Science & IT)

Q.53

Consider two set-associative cache memory architectures: WBC, which uses the write back policy, and WTC, which uses the write through policy. Both of them use the LRU (Least Recently Used) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

- (A) A read miss in WBC never evicts a dirty block
- (B) A read miss in WTC never triggers a write back operation of a cache block to main memory
- (C) A write hit in WBC can modify the value of the dirty bit of a cache block
- (D) A write miss in WTC always writes the victim cache block to main memory before loading the missed block to the cache

Question 12 of 55 | GATE 2024 Set 1 (CS1 S5) | Q.57 (Computer Science & IT)

Q.57

Consider the following code snippet using the fork() and wait() system calls.

Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
    fork();
    printf("hello");
    wait(NULL);
    x--;
}
```

The total number of times the printf statement is executed is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.24 (Computer Science & IT)

Q.24

Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

- (A) Allocate a new page table for a newly created process
 - (B) Translate a virtual address to a physical address using the page table
 - (C) Raise a trap when a virtual address is not found in the page table
 - (D) Raise a trap when a process tries to write to a page marked with read-only permission in the page table
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Question 14 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.25 (Computer Science & IT)

Q.25

Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?

- (A) P makes a blocking system call to read a block of data from the disk
- (B) P tries to access a page that is in the swap space, triggering a page fault
- (C) An interrupt is raised by the disk to deliver data requested by some other process
- (D) A timer interrupt is raised by the hardware

Question 15 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.26 (Computer Science & IT)

Q.26

Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?

- (A) Sorted
 - (B) Heap
 - (C) Unclustered tree index
 - (D) Unclustered hash index
- Organizing Institute: IISc, Bengaluru

Question 16 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.27 (Computer Science & IT)

Q.27

Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

- (A) 2PL permits only serializable schedules
- (B) With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
- (C) With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
- (D) A deadlock is possible with 2PL

Question 17 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.37 (Computer Science & IT)

Q.37

Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A (0, 10), B (2, 6), C (4, 3), and D (6, 7).

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

- (A) SRTF = 6, NP-SJF = 7
- (B) SRTF = 6, NP-SJF = 7.5
- (C) SRTF = 7, NP-SJF = 7.5
- (D) SRTF = 7, NP-SJF = 8.5

Question 18 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.46 (Computer Science & IT)

Q.46

Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also

share a global variable x (initialized to 0). The threads execute the code shown below.

```
// code of T1
```

```
wait(s1);
```

```
x = x+1;
```

```
print(x);
```

```
wait(s2);
```

```
signal(s1);
```

```
// code of T2
```

```
wait(s1);
```

```
x = x+1;
```

```
print(x);
```

```
signal(s2);
```

```
signal(s1);
```

Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

- (A) T1 runs first and prints 1, T2 runs next and prints 2
- (B) T2 runs first and prints 1, T1 runs next and prints 2
- (C) T1 runs first and prints 1, T2 does not print anything (deadlock)
- (D) T2 runs first and prints 1, T1 does not print anything (deadlock)

Question 19 of 55 | GATE 2024 Set 2 (CS2 S6) | Q.64 (Computer Science & IT)

Q.64

Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 210 bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X+Y$ is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 55 | GATE 2023 Single Session | Q.21 (Computer Science & IT)

Q.21

Which one or more of the following need to be saved on a context switch from one

thread (T1) of a process to another thread (T2) of the same process?

- (A) Page table base register
- (B) Stack pointer
- (C) Program counter
- (D) General purpose registers CS

Question 21 of 55 | GATE 2023 Single Session | Q.22 (Computer Science & IT)

Q.22

Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?

- (A) Function Call
- (B) malloc Call
- (C) Page Fault
- (D) System Call CS

Question 22 of 55 | GATE 2023 Single Session | Q.26 (Computer Science & IT)

Q.26

Which one or more of the following CPU scheduling algorithms can potentially cause starvation?

- (A) First-in First-Out
- (B) Round Robin
- (C) Priority Scheduling
- (D) Shortest Job First CS

Question 23 of 55 | GATE 2023 Single Session | Q.37 (Computer Science & IT)

Q.37

CS

Consider the two functions incr and decr shown below.

```
incr(){  
    wait(s);  
    X = X+1;  
    signal(s);  
}  
decr(){  
    wait(s);  
    X = X-1;
```

```
signal(s);  
}
```

There are 5 threads each invoking `incr` once, and 3 threads each invoking `decr` once, on the same shared variable `X`. The initial value of `X` is 10.

Suppose there are two implementations of the semaphore `s`, as follows:

I-1: `s` is a binary semaphore initialized to 1.

I-2: `s` is a counting semaphore initialized to 2.

Let `V1`, `V2` be the values of `X` at the end of execution of all the threads with implementations I-1, I-2, respectively.

Which one of the following choices corresponds to the minimum possible values of `V1`, `V2`, respectively?

- (A) 15, 7
- (B) 7, 7
- (C) 12, 7
- (D) 12, 8 CS

Question 24 of 55 | GATE 2023 Single Session | Q.51 (Computer Science & IT)

Q.51

Suppose in a web browser, you click on the `www.gate-2023.in` URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let `RTT` denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting `www.gate-2023.in` is also equal to `RTT`. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

- (A) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- (B) 5 RTTs, in case of persistent HTTP with pipelining.
- (C) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- (D) 6 RTTs, in case of persistent HTTP with pipelining. CS

Question 25 of 55 | GATE 2023 Single Session | Q.56 (Computer Science & IT)

Q.56

Consider the following two-dimensional array D in the C programming language, which is stored in row-major order:

```
int D[128][128];
```

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet:

```
for (int i = 0; i < 128; i++)
  for (int j = 0; j < 128; j++)
    D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 26 of 55 | GATE 2023 Single Session | Q.57 (Computer Science & IT)

Q.57

Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes.

The value of L is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 27 of 55 | GATE 2022 Single Session | Q.15 (Computer Science & IT)

Q.15

Consider the problem of reversing a singly linked list. To take an example, given the linked list below,

the reversed linked list should look like

Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in

(1)

O

space?

- (A) The best algorithm for the problem takes $(\) n \square$ time in the worst case.
- (B) The best algorithm for the problem takes $(\log) n n \square$ time in the worst case.
- (C) The best algorithm for the problem takes $2(\) n \square$ time in the worst case.

(D) It is not possible to reverse a singly linked list in (1) O space.

Question 28 of 55 | GATE 2022 Single Session | Q.19 (Computer Science & IT)

Q.19

Consider the following threads, T1, T2, and T3 executing on a single processor, synchronized using three binary semaphore variables, S1, S2, and S3, operated upon using standard wait() and signal(). The threads can be context switched in any order and at any time.

T1

T2

T3

```
while(true){
wait(S3);
print("C");
signal(S2); }
while(true){
wait(S1);
print("B");
signal(S3); }
while(true){
wait(S2);
print("A");
signal(S1); }
```

Which initialization of the semaphores would print the sequence BCABCABCA....?

- (A) S1 = 1; S2 = 1; S3 = 1
- (B) S1 = 1; S2 = 1; S3 = 0
- (C) S1 = 1; S2 = 0; S3 = 0
- (D) S1 = 0; S2 = 1; S3 = 1

Question 29 of 55 | GATE 2022 Single Session | Q.24 (Computer Science & IT)

Q.24

Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?

- (A) Each cache block in WB and WT has a dirty bit.
- (B) Every write hit in WB leads to a data transfer from cache to main memory.
- (C) Eviction of a block from WT will not lead to data transfer from cache to main memory.
- (D) A read miss in WB will never lead to eviction of a dirty block from WB.

Question 30 of 55 | GATE 2022 Single Session | Q.26 (Computer Science & IT)

Q.26

Which of the following statements is/are TRUE with respect to deadlocks?

- (A) Circular wait is a necessary condition for the formation of deadlock.
- (B) In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.
- (C) If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur.
- (D) In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in deadlock state.

Question 31 of 55 | GATE 2022 Single Session | Q.38 (Computer Science & IT)

Q.38

Which one of the following statements is FALSE?

- (A) The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
- (B) If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
- (C) The memory access time using a given inverted page table is always same for all incoming virtual addresses.
- (D) In a system that uses hashed page tables, if two distinct virtual addresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.

Question 32 of 55 | GATE 2022 Single Session | Q.42 (Computer Science & IT)

Q.42

Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

- (A) $P = 4, Q = 10, R = 6, S = 2$
- (B) $P = 2, Q = 9, R = 5, S = 1$
- (C) $P = 4, Q = 12, R = 5, S = 4$
- (D) $P = 3, Q = 7, R = 7, S = 3$

Question 33 of 55 | GATE 2022 Single Session | Q.64 (Computer Science & IT)

Q.64

Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string

7,2,7,3,2,5,3,4,6,7,7,1,5,6,1

the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 55 | GATE 2021 Set 1 (CS1) | Q.4 (Computer Science & IT)

Q.4

A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

- (A)
- (B)
- (C)
- (D)

Question 35 of 55 | GATE 2021 Set 2 (CS2) | Q.2 (Computer Science & IT)

Q.2

A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

- (A)
- (B)
- (C)
- (D)

Question 36 of 55 | GATE 2019 Single Session | Q.17 (Computer Science & IT)

Q.17 The following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main()
{
    int i;
    for (i=0; i<10; i++)
        if (i%2 == 0) fork();
    return 0;
```

}

The total number of child processes created is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 37 of 55 | GATE 2019 Single Session | Q.33 (Computer Science & IT)

Q.33 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

- (A) 16×2^{10}
- (B) 256×2^{10}
- (C) 4×2^{20}
- (D) 8×2^{20}

Question 38 of 55 | GATE 2019 Single Session | Q.41 (Computer Science & IT)

Q.41 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process

P1

P2

P3

P4

Arrival time

0

1

3

4

CPU burst time

3

1

3

Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 39 of 55 | GATE 2019 Single Session | Q.42 (Computer Science & IT)

Q.42 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

CS

11/15

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 40 of 55 | GATE 2018 Single Session | Q.10 (Computer Science & IT)

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- (A) $(D - M) / (X - M)$
- (B) $(X - M) / (D - M)$
- (C) $(D - X) / (D - M)$
- (D) $(X - M) / (D - X)$

Question 41 of 55 | GATE 2018 Single Session | Q.24 (Computer Science & IT)

Q.24 Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 42 of 55 | GATE 2018 Single Session | Q.39 (Computer Science & IT)

Q.39 In a system, there are three types of resources: E, F and G. Four processes P_0 , P_1 , P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $Max[P_2, F]$ is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

Allocation

Max

E F G

E

F G
 P0
 1 0
 1
 P0
 4
 3
 1
 P1 1 1
 2
 P1
 2
 1
 4
 P2 1 0
 3
 P2
 1
 3
 3
 P3 2 0
 0
 P3
 5
 4
 1

From the perspective of deadlock avoidance, which one of the following is true?

- (A) The system is in safe state.
- (B) The system is not in safe state, but would be safe if one more instance of E were available
- (C) The system is not in safe state, but would be safe if one more instance of F were available
- (D) The system is not in safe state, but would be safe if one more instance of G were available CS 14/20

Question 43 of 55 | GATE 2018 Single Session | Q.40 (Computer Science & IT)

Q.40 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N. Three semaphores empty, full and mutex are defined with respective initial values of 0, N and 1. Semaphore empty denotes the number of available slots in the buffer, for the consumer to read from. Semaphore full denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P, Q, R, and S, in the code below can be assigned either empty or full. The valid semaphore operations are: wait() and signal().

Producer:

Consumer:

```
do{
wait(P);
wait(mutex);
//Add item to buffer
signal(mutex);
signal(Q);
}while(1);
do{
wait(R);
wait(mutex);
//Consume item from buffer
signal(mutex);
signal(S);
}while(1);
```

Which one of the following assignments to P, Q, R and S will yield the correct solution?

- (A) P: full, Q: full, R: empty, S: empty
- (B) P: empty, Q: empty, R: full, S: full
- (C) P: full, Q: empty, R: empty, S: full
- (D) P: empty, Q: full, R: full, S: empty

Question 44 of 55 | GATE 2018 Single Session | Q.53 (Computer Science & IT)

Q.53 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ..., 199), and 256 sectors per track (numbered as 0, 1, ..., 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 45 of 55 | GATE 2016 Set 1 (CS1) | Q.22 (Computer Science & IT)

Q.22

Which one of the following is NOT a part of the ACID properties of database transactions?

- (A) Atomicity
- (B) Consistency
- (C) Isolation
- (D) Deadlock-freedom CS(Set A) 7/17 GATE 2016

Question 46 of 55 | GATE 2016 Set 1 (CS1) | Q.33 (Computer Science & IT)

Q.33

Consider a carry lookahead adder for adding two n -bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

- (A) $\Theta(1)$
- (B) $\Theta(\log(n))$
- (C) $\Theta(\sqrt{n})$
- (D) $\Theta(n)$ CS(Set A) 10/17 GATE 2016

Question 47 of 55 | GATE 2016 Set 1 (CS1) | Q.47 (Computer Science & IT)

Q.47

Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is megabytes.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 48 of 55 | GATE 2016 Set 1 (CS1) | Q.48 (Computer Science & IT)

Q.48

Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 49 of 55 | GATE 2016 Set 1 (CS1) | Q.49 (Computer Science & IT)

Q.49

Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct virtual page number.

The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is

CS(Set A)

15/17

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 50 of 55 | GATE 2016 Set 1 (CS1) | Q.50 (Computer Science & IT)

Q.50

Consider the following proposed solution for the critical section problem.

There are n

processes: $P_0 \dots P_{n-1}$. In the code, function pmax returns an integer not smaller than any of its arguments. For all i , $t[i]$ is initialized to zero.

Code for P_i :

```
do {
  c[i]=1; t[i] = pmax(t[0],...,t[n-1])+1; c[i]=0;
  for every j = i in {0,...,n-1} {
    while (c[j]);
    while (t[j] != 0 && t[j] <= t[i]);
  }
  Critical Section;
  t[i]=0;
  Remainder Section;
} while (true);
```

Critical Section;

$t[i]=0$;

Remainder Section;

} while (true);

Which one of the following is TRUE about the above solution?

- (A) At most one process can be in the critical section at any time
- (B) The bounded wait condition is satisfied
- (C) The progress condition is satisfied
- (D) It cannot cause a deadlock

Question 51 of 55 | GATE 2016 Set 1 (CS1) | Q.51 (Computer Science & IT)

Q.51

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \dots, O_k\}$. This is done in the following manner:

Step 1. T acquires exclusive locks to $O_1, \dots, O_k$ in increasing order of their addresses.

Step 2. The required operations are performed.

Step 3. All locks are released.

This protocol will

- (A) guarantee serializability and deadlock-freedom
- (B) guarantee neither serializability nor deadlock-freedom
- (C) guarantee serializability but not deadlock-freedom
- (D) guarantee deadlock-freedom but not serializability CS(Set A) 16/17 GATE 2016

Question 52 of 55 | GATE 2016 Set 2 (CS2) | Q.20 (Computer Science & IT)

Q.20

In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

- (A) LRU (Least Recently Used)
- (B) OPT (Optimal Page Replacement)
- (C) MRU (Most Recently Used)
- (D) FIFO (First In First Out) CS(Set B) 5/18 GATE 2016

Question 53 of 55 | GATE 2016 Set 2 (CS2) | Q.47 (Computer Science & IT)

Q.47

Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process

Arrival Time

Burst Time

P1

0

10

P2

3

6

P3

7

1

P4

8

3

The average turn around time of these processes is milliseconds.

CS(Set B)

14/18

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 54 of 55 | GATE 2016 Set 2 (CS2) | Q.48 (Computer Science & IT)

Q.48

Consider the following two-process synchronization solution.

Process 0

Entry: loop while (turn == 1);

(critical section)

Exit:

turn = 1;

Process 1

Entry: loop while (turn == 0);

(critical section)

Exit:

turn = 0;

The shared variable turn is initialized to zero. Which one of the following is TRUE?

- (A) This is a correct two-process synchronization solution.
- (B) This solution violates mutual exclusion requirement.
- (C) This solution violates progress requirement.
- (D) This solution violates bounded wait requirement.

Question 55 of 55 | GATE 2016 Set 2 (CS2) | Q.49 (Computer Science & IT)

Q.49

Consider a non-negative counting semaphore S. The operation P(S) decrements S, and V(S) increments S. During an execution, 20 P(S) operations and 12 V(S) operations are issued in some order. The largest initial value of S for which at least one P(S) operation will remain blocked is

.

CS(Set B)

15/18

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]